

15.4 Laplacian Matrix and Spectral Clustering



The core skills you'll master in this section:

- ▶ The basic workflow of spectral clustering.
- ▶ An unweighted undirected graph's adjacency matrix is symmetric, with 1 indicating a connection and 0 indicating no connection.
- ▶ An undirected graph's degree matrix is diagonal, with each diagonal element equal to that node's degree, reflecting how active the node is.
- ▶ Nodes with high degree are typically the central figures in a social network, while nodes with low degree tend to sit on the margins.
- ▶ The Laplacian matrix is obtained by subtracting the adjacency matrix from the degree matrix, and it captures the graph's connectivity structure.
- ▶ Normalization removes the effect of differing node degrees, preventing highly connected nodes from dominating the clustering result.
- ▶ Selecting the eigenvectors corresponding to the smallest eigenvalues, nodes that are close together in the resulting low-dimensional space represent tight social relationships.

This section introduces how to use spectral clustering to analyze a social network. The core element in spectral clustering is the Laplacian matrix, which is constructed from the degree matrix and adjacency matrix introduced earlier. The key computational tool is eigenvalue decomposition, which you're already familiar with by now.

Social Network

A friendship graph in human social interaction is a typical undirected graph — nodes represent people, and edges represent friendships between two people. The karate club membership graph is a classic undirected graph that describes the social relationships among a group of club members.

In Figure 1, this undirected graph — nodes represent members of the club; edges represent relationships between two members, for example, whether they frequently train or socialize together. The numbers in the graph are the club members' ID numbers.

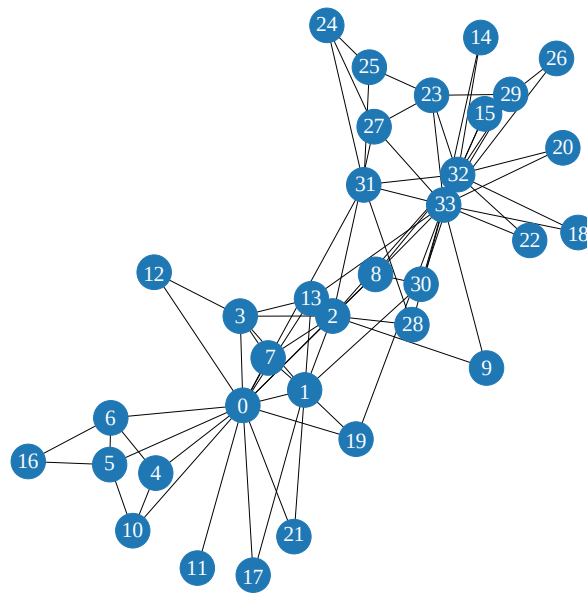


Figure 1. The Karate Club Membership Undirected Graph

This is a famous dataset commonly used to study network structure, introduced by Zachary in a 1977 study; this undirected graph is therefore known as Zachary's Karate Club.

It was originally used to analyze social dynamics and factional splits within a group. In Zachary's study, due to a conflict between the club's leaders, the members eventually split into two factions. This kind of split can be automatically identified using community-detection algorithms on the undirected graph.

Spectral Clustering

Next, we'll use spectral clustering to cluster the nodes in Figure 1.

Clustering is a method of dividing data into several groups, or clusters, based on similarity. In clustering, data within the same cluster should be as similar to each other as possible, while data in different clusters should be as different as possible.

In simple terms, clustering means grouping similar things together and separating dissimilar things.

The basic workflow of spectral clustering is as follows:

- ▶ First, given a dataset or a graph structure, compute its undirected graph's adjacency matrix.
- ▶ Next, compute the degree matrix.
- ▶ Then, using the degree matrix and adjacency matrix, compute the graph's Laplacian matrix.
- ▶ Compute the normalized Laplacian matrix.
- ▶ Perform eigenvalue decomposition on the normalized Laplacian matrix.
- ▶ Arrange the corresponding eigenvectors in ascending order of eigenvalue, and select the eigenvectors corresponding to the first few smallest eigenvalues to construct a low-dimensional space.
- ▶ Finally, cluster the nodes within this low-dimensional space.

Let's now follow this approach to walk through the rest of this section.

Adjacency Matrix

Based on the connections between nodes in Figure 1, this undirected graph, we first compute the adjacency matrix shown in Figure 2, namely A .

As mentioned earlier in this chapter, an undirected graph's adjacency matrix A is symmetric.

Especially for an unweighted undirected graph, an element of adjacency matrix A equal to 0 means no edge exists between the nodes; an element equal to 1 means an edge exists between the nodes.

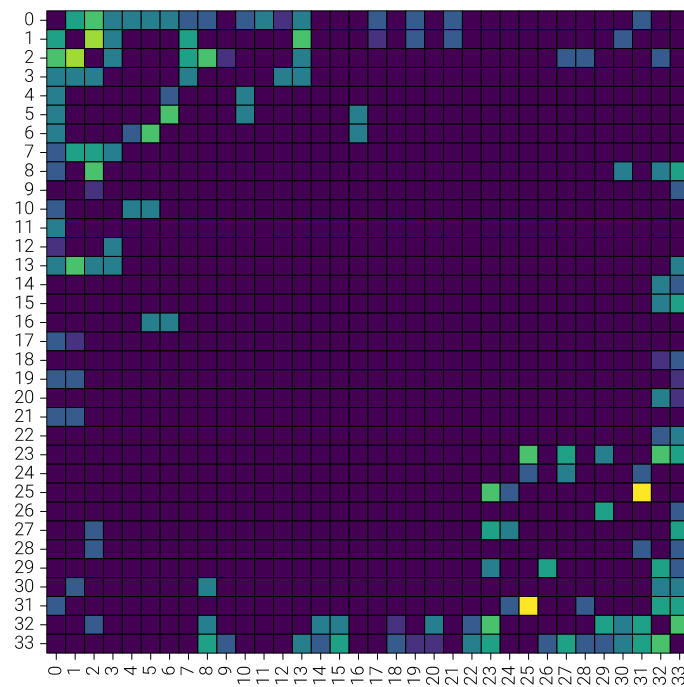


Figure 2. Adjacency Matrix

Degree Matrix

Next, compute the degree matrix D .

As shown in Figure 3, the undirected graph's degree matrix D is a diagonal matrix, where each diagonal element is the degree of the corresponding node — that is, the total number of edges connected to that node.

? Please review the relationship between an undirected graph's degree matrix and adjacency matrix, and think about how to compute the degree matrix from the adjacency matrix in Python.

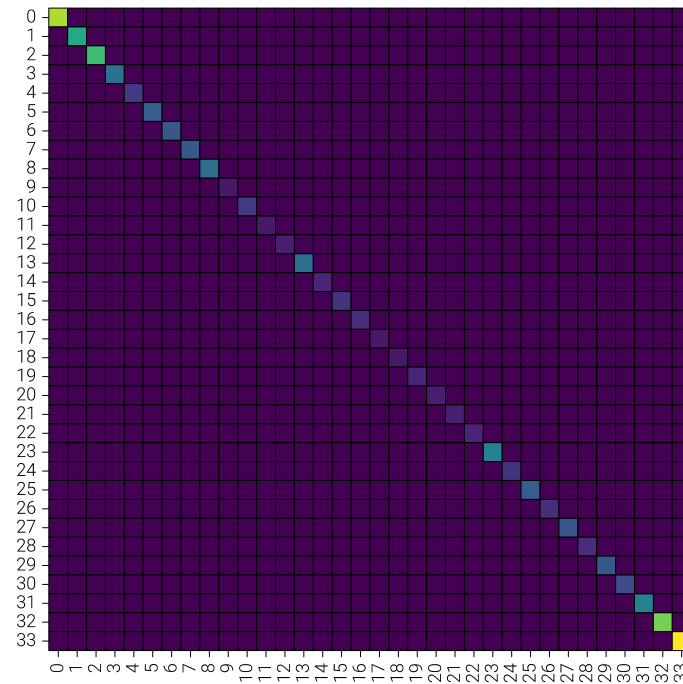


Figure 3. Degree Matrix

A node's degree is the number of edges connected to it. The higher a node's degree, the more active and important that node is in the network — it has direct connections with more other nodes, and it typically plays a key role in information flow, resource allocation, or social networks.

Figure 4 shows a bar chart of each node's degree. It's easy to see that nodes 0 and 33 have especially high degrees — they're effectively the "central figures" of the group.

Conversely, the lower a node's degree, the fewer connections it has in the network — it may play a peripheral role, participate less, or sit in a structurally isolated position.

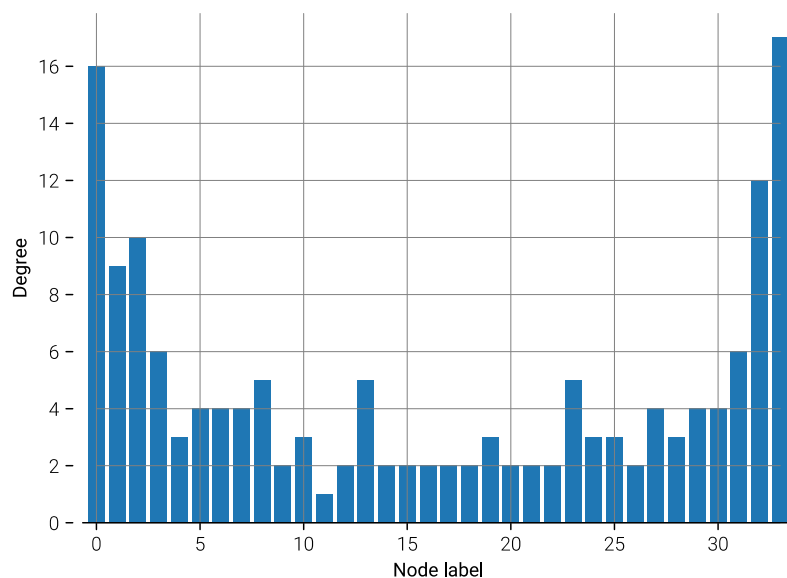


Figure 4. The Degree of Each Node

Overall, degree reflects a node's direct influence and level of participation in the network, and is an important measure of a node's status. Figure 5 shows the nodes rendered by degree — and it looks like we've already spotted two "friend circles."

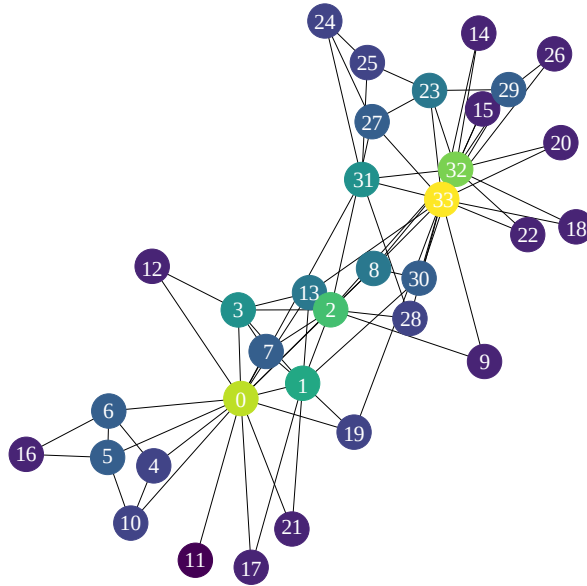


Figure 5. Nodes Rendered by Degree

Laplacian Matrix

Then, based on the degree matrix $\mathbf{D}$ and adjacency matrix $\mathbf{A}$, compute the graph's Laplacian matrix $\mathbf{L}$

$$\mathbf{L} = \mathbf{D} - \mathbf{A}$$

For an undirected graph, both the degree matrix $\mathbf{D}$ and adjacency matrix $\mathbf{A}$ are symmetric, so the Laplacian matrix $\mathbf{L}$ is clearly symmetric too, as specifically shown in Figure 6.

The Laplacian matrix is an important representation of a graph's structure, capturing the connectivity between nodes and its overall structural properties. The Laplacian matrix's eigenvalues and eigenvectors can also reveal the partition structure within a network.

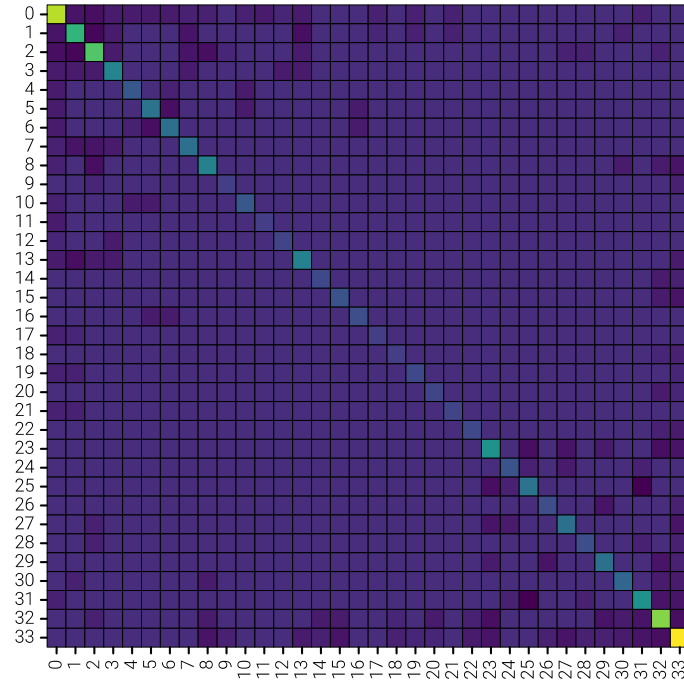


Figure 6. Laplacian Matrix

Normalized Laplacian Matrix

To further eliminate the effect of node degree on the computation, we typically need to compute the (symmetrically) normalized Laplacian matrix

$$\mathbf{L}_{\text{normalized}} = \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2} = \mathbf{D}^{-1/2} (\mathbf{D} - \mathbf{A}) \mathbf{D}^{-1/2}$$

That is, multiply both sides of the Laplacian matrix $\mathbf{L}$ by the negative square root of the degree matrix $\mathbf{D}$ — this is equivalent to scaling $\mathbf{L}$'s rows and columns respectively.

As shown in Figure 7, the main-diagonal elements of the normalized Laplacian matrix are all 1.

In graphs where node degrees vary widely, the original (unnormalized) Laplacian matrix tends to let highly connected nodes dominate the overall structure, leading to imbalanced clustering or partitioning results. After normalization, each node's connection strength is effectively standardized, putting all nodes on more equal footing in the computation.

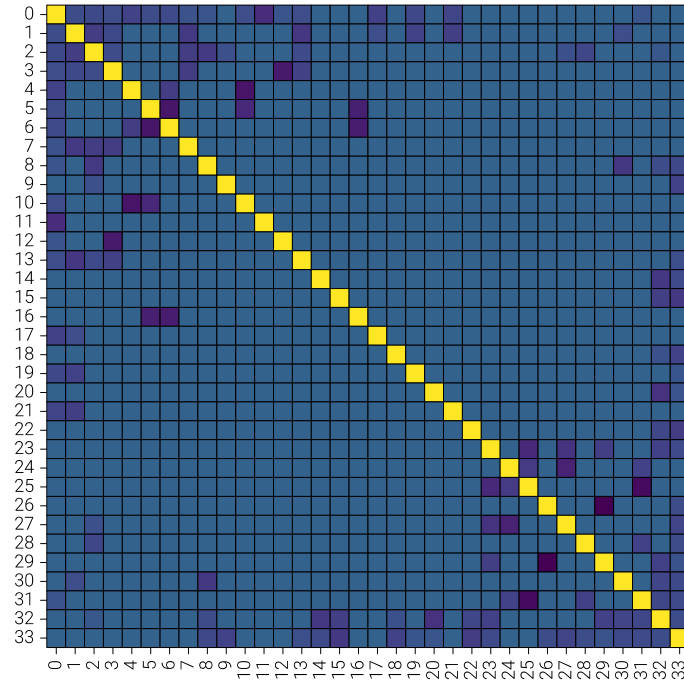


Figure 7. Normalized Laplacian Matrix

Eigenvalue Decomposition

Once we have the normalized Laplacian matrix, we perform eigenvalue decomposition,

$$\mathbf{L}_{\text{normalized}} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$$

The result is shown in Figure 8.

Since the normalized Laplacian matrix is a real symmetric matrix, Figure 8 is a spectral decomposition.

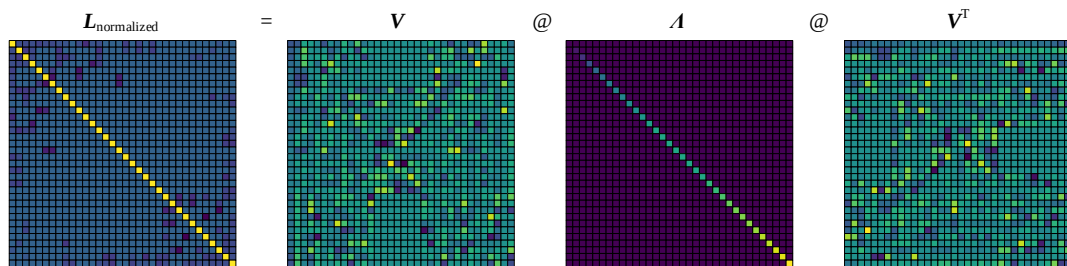


Figure 8. The Eigenvalue Decomposition (Spectral Decomposition) of the Normalized Laplacian Matrix

Based on the size of the eigenvalues, the corresponding eigenvectors are arranged in ascending order, as shown in Figure 9 and Figure 10.

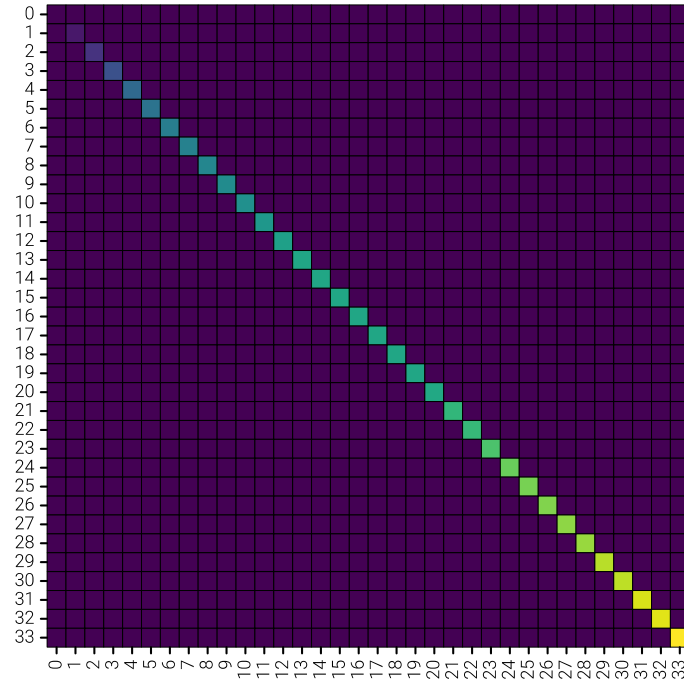


Figure 9. The Diagonal Square Matrix of Eigenvalues

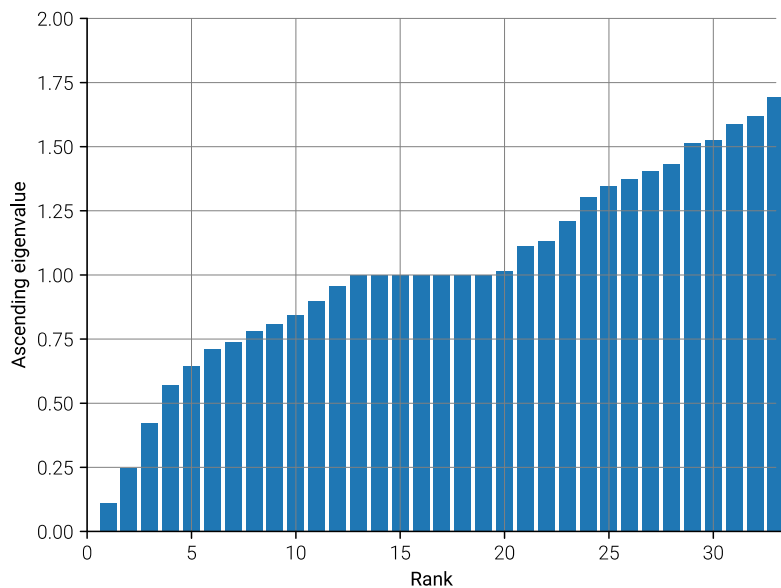


Figure 10. Eigenvalues Arranged in Increasing Order

Because Figure 1's undirected graph is relatively simple, we select the eigenvectors corresponding to the first 2 smallest eigenvalues (Figure 11's orthogonal matrix's first two columns, $\mathbf{v}_1$ and $\mathbf{v}_2$), and then use these eigenvectors to represent membership in a new low-dimensional space, as shown in Figure 12.

Figure 12 – the red dots have a y-coordinate greater than 0, and the blue dots have a y-coordinate less than 0. In effect, we've already split the members into two clusters, as specifically shown in Figure 13.

The normalized Laplacian matrix essentially captures the similarity between nodes. After spectral decomposition, the eigenvectors map each node into a new low-dimensional space. In this space,

nodes that are closely connected end up close together, while sparsely connected nodes end up farther apart.

In principal component analysis, we choose the projection direction corresponding to the largest eigenvalue because, along that direction, the data is spread out the most and carries the most information. Spectral clustering, by contrast, aims for aggregation — the direction it finds is one that brings closely connected nodes even closer together, making it easier to form clusters.

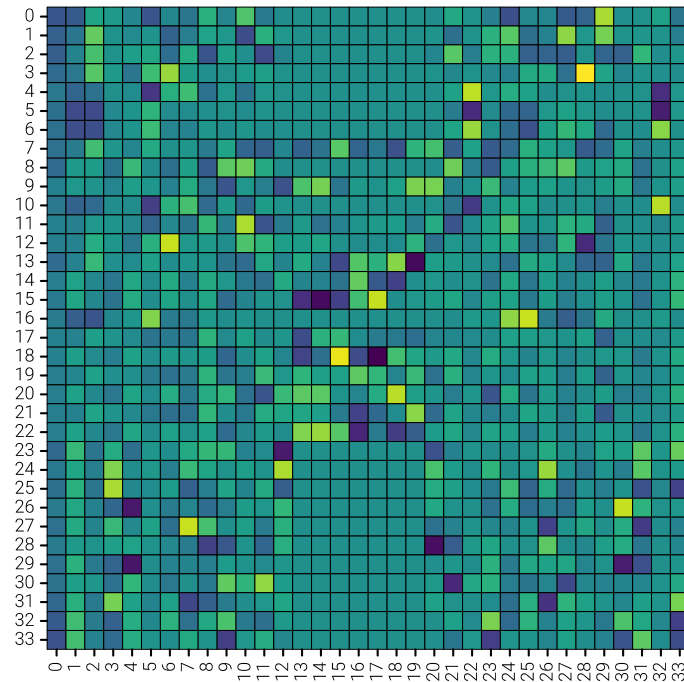


Figure 11. Orthogonal Matrix V

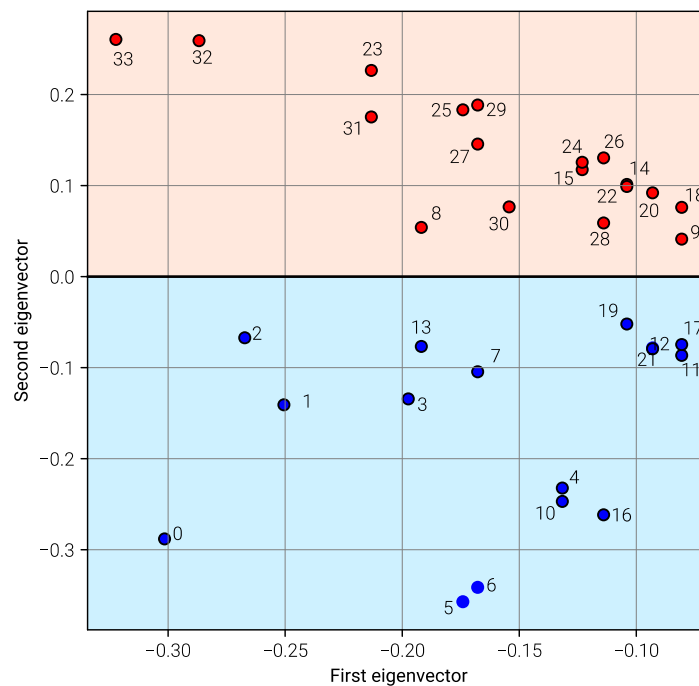


Figure 12. The Scatter Plot Formed by the First Two Eigenvectors

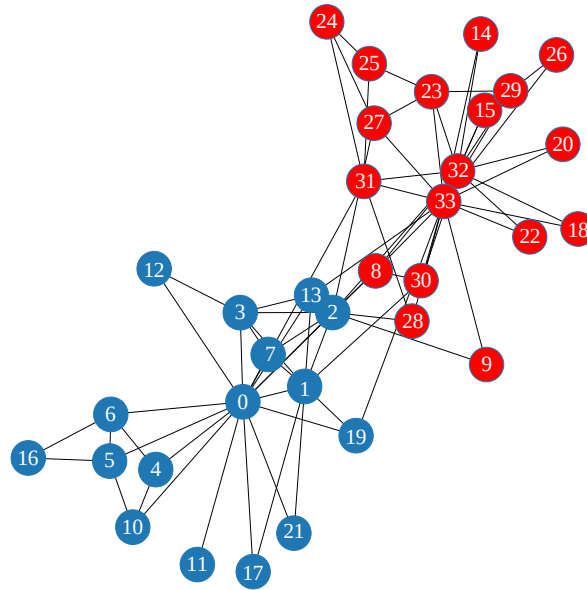


Figure 13. Partitioning the Undirected Graph



LA_15_04_01.ipynb performs all the computations and visualizations for this section's spectral decomposition.

Epilogue

Making linear algebra genuinely understood and well applied is the goal of Linear Algebra Made Easy and Visual. With the help of Python programming, geometric thinking, and Manim math animations, we hope Linear Algebra Made Easy and Visual truly makes linear algebra "not hard" at all!

Learning linear algebra shouldn't be a dry, tedious numbers game, and it certainly shouldn't be a painful slog of memorizing formulas by rote. Learning linear algebra should be a visual, geometric feast — an expedition deep into the jungle of numbers.

Linear Algebra Made Easy and Visual hopes to help readers open a window onto the rich geometric intuition, practical value, and even mathematical beauty behind linear algebra. Truly mastering linear algebra means being able to view data, models, phenomena, and the world from a higher vantage point.

Of course, this book is only a starting point. The world of linear algebra is vast and deep. We hope this book plants a seed in your mind, so that in your future studies and work, you'll keep exploring and keep growing.

The beauty of mathematics doesn't lie in complexity, but in seeing through abstraction to the order and essence beneath it. May every reader, free and confident, carrying a burning curiosity, turn linear algebra into an inexhaustible source of strength for exploring the world.